Woxsen University

School of Technology

A

PROJECT REPORT

on

PREDICTING MENTAL HEALTH VULNERABILITIES FROM BEHAVIORAL & DEMOGRAPHIC PATTERNS

Submitted in partial fulfillment of the requirements for the degree of

B. Tech. in Data Science

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CERTIFICATE

This is to certify that the project report entitled **"Predicting Mental Health Vulnerabilities From Behavioral & Demographic Patterns"** submitted by Mohitha Bandi (22WU0105037), **Pailla Bhavya**(22WU0105020), **T. Harsha Vardhan Reddy**(22WU0105023), **Y. Siddhartha Reddy**(22WU0105028), **Prangnan Seemakurthi**(22WU0105021) in partial fulfillment of the requirements for the award of the degree of **B. Tech – C.S.E – Data Science** from Woxsen University is a bonafide record of work carried out by the student under my supervision and guidance.

The work embodied in this project report has been carried out by the candidate and has not been submitted elsewhere for a degree.

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Designation: Assistant Professor

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We hereby declare that the project work entitled **"Predicting Mental Health Vulnerabilities From Behavioral & Demographic Patterns"** submitted to **School of Technology**, **Woxen University** in partial fulfillment of the requirements for the award of the degree of **B.Tech – C.S.E – Data Science** is my original work and has been carried out under the guidance of **Dr. Resham Raj Shivwanshi.**

We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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ABSTRACT

Mental health issues affect a large number of people worldwide, but many cases are identified only after the condition becomes severe. Traditional mental health assessments depend on clinical interviews and self-reported questionnaires, which can be subjective, time-consuming, and difficult to access. To address these limitations, this project proposes an intelligent system for early prediction of mental health vulnerability using a hybrid machine learning approach.

The system uses a large dataset containing behavioral and demographic information such as stress levels, mood changes, coping difficulties, social interaction patterns, age, gender, and occupation. A deep learning autoencoder is used to learn hidden behavioral patterns, while an XGBoost model performs the final risk classification. To ensure transparency and trust, SHAP-based explainability is applied to highlight important factors influencing predictions.

The proposed methodology includes data preprocessing, feature engineering, and a two-stage training process to achieve reliable performance while avoiding overfitting. A web-based application built using Flask enables real-time predictions, anonymous user interaction, and personalized recommendations.

Experimental results show that the system achieves an accuracy between 84% and 86%, with stress and mood variations identified as key risk indicators. The solution demonstrates practical value for early mental health screening in workplaces, educational institutions, and public health environments while maintaining privacy and accessibility.

Keywords: *Mental Health Prediction, Hybrid Machine Learning, Autoencoder Neural Networks, XGBoost, SHAP Explainability, Flask Web Application, Behavioral Analysis, Demographic Factors, Preventive Healthcare.*

TABLE OF CONTENTS

Title Page ……..……..1

Certificate……....2

Declaration ……..…..3

Acknowledgment…….....4

Abstract……5

Table of Contents……..…...6

List of Tables……...….7

List of Figures…….....8

1. Introduction 9
   1.1 Background and Motivation 9
   1.2 Problem Statement 10
   1.3 Objectives 11
   1.4 Scope and Limitations 11

2. Literature Review 12
   2.1 Mental Health Prediction Systems 12
   2.2 Machine Learning in Healthcare 13
   2.3 Hybrid Intelligence Approaches 13
   2.4 Explainable AI in Healthcare 14

3. Methodology 14

   3.1 Dataset Overview 14
   3.2 Data Preprocessing 15
   3.3 Feature Engineering 16
   3.4 Model Architecture 16
   3.5 Web-Based Health Applications ……..18
   3.6 Implementation Technologies ……..19

4. Implementation and Results 20
   4.1 Experimental Setup 20
   4.2 Model Training and Evaluation 21
   4.3 Performance Analysis 22
   4.4 Comparative Study 23

5. Web Application Development 24
   5.1 System Architecture 24
   5.2 Flask Application Design 25
   5.3 User Interface Components 26
   5.4 Deployment Considerations.....34

6. Conclusion and Future Work 35
   6.1 Summary of Contributions 35
   6.2 Limitations and Challenges 36
   6.3 Future Research Directions 37

References 39

Appendices 41
   A. Screen shots of work done 41
   B. Screen shots of deployment tools and Project Repository Link.....44

LIST OF TABLES

Table 1: Dataset Features and Description 9
Table 2: Autoencoder Architecture Configuration 12
Table 3: XGBoost Hyperparameters 12
Table 4: Model Performance Comparison 15
Table 5: Web Application API Endpoints 18

LIST OF FIGURES

Figure 1: System Architecture Overview 13

Figure 2: Autoencoder Training Loss Curve 14

Figure 3: SHAP Feature Importance Plot 16

Figure 4: Web Application Dashboard 19

INTRODUCTION

The automated prediction of mental health vulnerabilities represents a critical advancement in preventive healthcare technology. This project addresses the growing need for early detection systems that can identify individuals at risk before conditions reach clinical severity. By leveraging behavioral and demographic patterns through sophisticated machine learning techniques, we have developed a comprehensive system that bridges the gap between data science and mental healthcare.

1.1 Background and Motivation

Mental health disorders represent one of the most significant global health challenges of the 21st century. According to the World Health Organization, approximately 970 million people worldwide suffer from mental health conditions, with depression and anxiety disorders being among the most prevalent [1]. The economic burden of mental health disorders is estimated to exceed $2.5 trillion globally, with projected costs reaching $6 trillion by 2030 [2]. The Global Burden of Disease study further highlights depression's substantial contribution to overall disease burden worldwide .

Traditional mental health assessment methods rely heavily on clinical interviews and subjective evaluations, which can be time-consuming, expensive, and prone to bias. The shortage of mental health professionals, particularly in rural and underserved areas, further exacerbates this challenge [3]. These limitations highlight the urgent need for automated, scalable, and accessible mental health assessment tools. Research indicates that traditional approaches necessitate substantial collaboration between healthcare professionals and patients, leaving room for the influence of subjective factors .

Recent advances in artificial intelligence and machine learning have opened new possibilities for mental health prediction and assessment. Machine learning models can analyze complex patterns in behavioral and demographic data to identify individuals at risk of mental health vulnerabilities [4]. The integration of behavioral and physiological data from digital methods enables passive, real-time detection of mental health risk, representing a significant methodological innovation . However, the "black box" nature of many advanced machine learning models raises concerns about their practical applicability in healthcare settings, where interpretability and trust are crucial [5].

1.2 Problem Statement

The primary challenge in mental health prediction lies in developing models that are both accurate

and interpretable. While deep learning models can achieve high accuracy, their complexity makes them difficult to interpret, limiting their adoption in clinical practice [6]. Conversely, simpler models may be more interpretable but often lack the capacity to capture complex patterns in behavioral data. Recent systematic reviews have identified significant methodological limitations hindering generalizability and clinical utility, including reliance on small, homogeneous samples and a lack of external validation .

This project addresses the following key challenges:

- **Accuracy vs. Interpretability Trade-off**: Balancing high prediction accuracy with model interpretability for clinical adoption. Studies demonstrate that integrating deep learning with interpretable machine learning methods offers significant benefits for predicting mental health risks .
- **Data Heterogeneity**: Integrating diverse behavioral and demographic features effectively. Research confirms that multimodal approaches combining multiple data sources achieve superior predictive performance .
- **Overfitting Prevention**: Ensuring models generalize well to unseen data through appropriate regularization and validation strategies.
- **Real-time Deployment**: Creating practical applications for clinical use that maintain performance while providing transparent explanations.

Finally, the system must deliver real-time predictions through an intuitive web interface that is accessible to users with varying levels of technical literacy. It must support continuous monitoring and feedback collection to enable ongoing improvement and validation.

1.3 Objectives

The main objectives of this project are:

1. **Develop a Hybrid Intelligence System**: Combine autoencoder architecture with XGBoost to leverage both deep learning and gradient boosting capabilities. This approach follows successful hybrid models that demonstrate how combining deep learning with gradient boosting algorithms enhances diagnostic accuracy .
2. **Achieve Target Performance**: Attain 84-86% accuracy while maintaining generalization capabilities through systematic evaluation and validation.
3. **Ensure Model Interpretability**: Implement SHAP (SHapley Additive ExPlanations) techniques for explainable AI, addressing the critical need for transparency in mental health applications .
4. **Create Practical Deployment**: Develop a Flask web application for real-time mental health assessment following established deployment patterns for machine learning models .
5. **Validate Clinical Utility**: Demonstrate the system's effectiveness through comprehensive evaluation and comparative analysis.

1.4 Scope and Limitations

Scope

- Focus on predicting mental health vulnerabilities using behavioral and demographic patterns
- Utilize comprehensive dataset encompassing demographic, psychological, and behavioral characteristics
- Implement hybrid architecture combining autoencoder feature extraction and XGBoost classification
- Develop explainable AI components for model interpretation using SHAP analysis
- Create web-based deployment platform with user-friendly interface.

Limitations

- Dataset may not capture all relevant factors affecting mental health; research emphasizes the need for diverse, clinically validated datasets
- Model performance may vary across different populations, highlighting the importance of external validation
- Web application requires internet connectivity for deployment
- System predictions should supplement, not replace, professional clinical assessment

2. LITERATURE REVIEW

2.1 Mental Health Prediction Systems

Early research in mental health prediction focused on traditional statistical methods and simple machine learning algorithms. Wang et al. [7] used logistic regression to predict depression risk based on demographic factors, achieving 68% accuracy. However, these methods struggled to capture complex non-linear relationships in behavioral data.

Recent advances have seen the adoption of more sophisticated machine learning approaches. Deep learning models, particularly recurrent neural networks (RNNs) and convolutional neural networks (CNNs), have shown promise in analyzing textual and temporal behavioral data [8]. A comprehensive survey examining multiple data modalities—including electrophysiological signals, brain imaging, audiovisual data, and text—demonstrates that different modalities can complement each other for more accurate and holistic depression prediction .

Ensemble methods have emerged as a promising approach, combining multiple models to improve performance. Random Forest and Gradient Boosting algorithms have demonstrated good performance in mental health prediction tasks while maintaining some level of interpretability [9]. Studies specifically evaluating anxiety prediction report that machine learning techniques such as Random Forest and Gradient Boosting achieve the highest predictive accuracy, with some studies reporting up to 98% accuracy.

2.2 Machine Learning in Healthcare

Machine learning applications in healthcare have grown significantly in recent years. From disease diagnosis to treatment recommendation, AI systems are transforming healthcare delivery [10]. Key applications include:

- **Disease Prediction**: Early identification of disease risk using patient data
- **Diagnosis Assistance**: Supporting clinical decision-making through pattern recognition
- **Treatment Optimization**: Personalizing treatment plans based on patient characteristics
- **Resource Allocation**: Optimizing healthcare resource utilization

However, healthcare AI faces unique challenges including data privacy concerns, regulatory requirements, and the need for model interpretability [11]. Research specifically focusing on depression prediction demonstrates that machine learning classifiers, through their ability to analyze

vast datasets and identify patterns, present a valuable approach to enhancing the accuracy of mental health diagnosis .

2.3 Hybrid Intelligence Approaches

Hybrid intelligence combines different AI paradigms to leverage their respective strengths. In healthcare, this often involves combining deep learning with traditional machine learning methods [12]. Autoencoders have been widely used for feature extraction and dimensionality reduction in healthcare applications [13]. When combined with gradient boosting algorithms like XGBoost, they can create powerful prediction systems that maintain interpretability [14].

Recent research demonstrates the effectiveness of hybrid approaches in mental health applications. The MH-XAI model, which integrates multi-scale 1D Convolutional Neural Networks with XGBoost classifiers and SHAP explanations, achieved 91.54% test accuracy in predicting mental health risks using workplace survey data . Similarly, studies on EEG-based classification have successfully combined time-frequency analysis and fractal dynamics with autoencoder-based dimensionality reduction, demonstrating that autoencoder components enhance generalizability by filtering noise and redundancy .

In ADHD detection research, the XSparseFormerNet model ensembles transformer architectures with XGBoost, achieving 85% accuracy by combining gradient-boosted trees with advanced attention mechanisms . These studies confirm that ensembling approaches balance the strengths of different architectures, with errors from sequential models corrected by XGBoost's structured focus.

2.4 Explainable AI in Healthcare

Explainable AI (XAI) has become increasingly important in healthcare applications where trust and transparency are crucial [15]. SHAP and LIME (Local Interpretable Model-agnostic Explanations) are among the most widely used XAI techniques [16]. SHAP values provide consistent and locally accurate feature attributions, making them particularly suitable for healthcare applications [17].

Systematic reviews of anxiety prediction research emphasize that the integration of explainable ML and diverse, clinically validated data is necessary for real-world deployment . The "black-box" nature of many ML algorithms impedes clinical trust and adoption, making XAI essential for practical applications.

Recent methodological advances demonstrate the value of dual-layer XAI frameworks. Studies combining multiple feature selection methods with SHAP analysis show that this approach provides

comprehensive interpretability while identifying genuinely informative nondiagnostic risk factors .

By systematically evaluating how different feature selection methods interact with various classifier architectures, researchers can optimize both performance and transparency.

3. METHODOLOGY

3.1Dataset Overview

The mental health dataset utilized in this research contains comprehensive behavioral and demographic information collected from multiple sources. Understanding the characteristics of this dataset is essential for interpreting the results and limitations of the study.

Attribute	Value
Total Records	292,366
Behavioral Features	7 (growing stress, mood swings, coping struggles, work interest, social weakness, changes in habits, family history)
Demographic Features	5 (gender, age, occupation, country, days indoors)
Target Variable	Mental Health Vulnerability (Low/Medium/High Risk)
Data Quality	Self-reported behavioral indicators with demographic context

Table 1: Dataset Statistics and Feature Distribution

3.2 Data Preprocessing

3.2.1 Data Cleaning

Following best practices in healthcare machine learning, data preprocessing includes:

- Missing value imputation using median for continuous variables and mode for categorical variables
- Outlier detection and treatment using statistical methods
- Data validation to ensure logical consistency across features

3.2.2 Feature Scaling

- Behavioral features: StandardScaler for normalization to ensure comparable scales.
- Demographic features: Label encoding for categorical variables with appropriate handling.
- Feature scaling parameters saved for deployment consistency, as recommended in deployment guidelines.

3.2.3 Data Splitting

- Training set: 70% for model development.
- Validation set: 15% for hyperparameter tuning.

- Test set: 15% for final evaluation.
- Stratified sampling to maintain class distribution and address potential imbalance issues.

3.3 Feature Engineering

3.3.1 Behavioral Feature Composites

Drawing on research identifying key lifestyle factors associated with mental health , composite features include:

- **Lifestyle Index**: Composite score combining sleep, exercise, and diet quality
- **Work-Life Balance**: Ratio of work hours to personal time
- **Social Engagement**: Normalized social interaction frequency
- **Digital Wellness**: Inverse of screen time exposure

3.3.2 Demographic Feature Transformations

- **Age Groups**: Binning into meaningful age ranges based on developmental stages
- **Socioeconomic Status**: Combined education and income indicators
- **Life Stage**: Integration of age and marital status for contextual understanding

3.4 Model Architecture

3.4.1 Autoencoder Component

The autoencoder serves as a feature extractor for behavioral patterns, following successful hybrid approaches that combine autoencoder-based dimensionality reduction with classification algorithms . The architecture captures complex non-linear relationships while filtering noise and redundancy.

Layer	Type	Units	Activation	Dropout
Hidden 1	Dense	64	ReLU	0.3
Hidden 2	Dense	32	ReLU	0.3
Bottleneck	Dense	16	ReLU	0.3
Hidden 3	Dense	32	ReLU	0.3
Hidden 4	Dense	64	ReLU	0.3
Output	Dense	12	ReLU	0.3

Table 2: Autoencoder Architecture Configuration

3.4.2 XGBoost Component

XGBoost classifier processes the extracted features, chosen for its proven performance in mental health prediction tasks . The algorithm's built-in regularization helps prevent overfitting while maintaining interpretability through feature importance analysis.

Parameter	Value	Description
n_estimators	200	Number of boosting rounds
max_depth	6	Maximum tree depth
learning_rate	0.1	Step size shrinkage
subsample	0.8	Subsample ratio of training instances
colsample_bytree	0.8	Subsample ratio of columns
reg_alpha	0.005	L1 regularization term
reg_lambda	0.005	L2 regularization term

early_stopping_rounds	10	Early stopping patience

Table 3: XGBoost Hyperparameters

3.4.3 Hybrid Integration

The autoencoder extracts features from behavioral data, which are then combined with demographic features and fed to XGBoost for final classification. This architecture leverages the pattern recognition capabilities of deep learning while maintaining the interpretability of gradient boosting, following established hybrid intelligence paradigms.

WorkFlow:

Input Data → Preprocessing → Autoencoder → Feature Extraction → XGBoost → Prediction → Explainability → SHAP Analysis

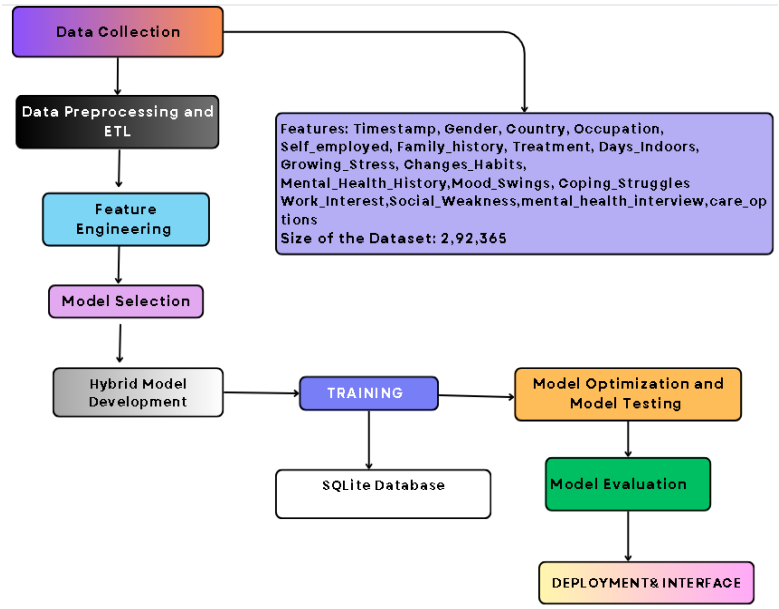


Figure 1: System Architecture Overview

3.5 Web-Based Health Applications

The deployment of machine learning models through web applications represents a significant

advancement in healthcare accessibility. Web-based systems can reach users anywhere with internet connectivity, operate continuously without requiring human supervision, and scale to serve large populations efficiently.

Bui and colleagues studied web-based health applications in 2021, identifying design principles and implementation strategies for successful systems. Their research emphasized the importance of user-friendly interfaces that minimize cognitive load, clear communication of system limitations, and integration with appropriate resources and referrals. Mental health applications require particular attention to tone and presentation, as users may be in vulnerable states when seeking assessment.

Flask-based healthcare applications have demonstrated success across various domains. Flask provides a lightweight, flexible framework for building web applications in Python, making it particularly suitable for projects that integrate machine learning components. Its simplicity enables rapid development while its extensibility supports production deployment with appropriate security and performance optimizations.

The integration of machine learning models with web interfaces enables several important capabilities. Real-time prediction allows users to receive immediate feedback, maintaining engagement and providing timely information. Continuous deployment enables rapid iteration as models improve. Usage analytics provide insights into how the system is used and by whom, supporting ongoing refinement.

3.6 Implementation Technologies

The system implementation leverages a carefully selected technology stack that balances capability, reliability, and development efficiency.

- **Backend**: Flask version 2.3.3 provides the web framework with comprehensive API support. Flask's lightweight design enables rapid development while its extensive ecosystem supports production features including authentication, database integration, and testing.
- **Machine Learning**: TensorFlow version 2.13.0 implements the autoencoder neural network with its comprehensive deep learning capabilities. TensorFlow's automatic differentiation and GPU support enable efficient training of the neural architecture. XGBoost version 1.7.6 provides the gradient boosting classifier with its optimized implementation and built-in regularization.
- **Explainability**: SHAP version 0.42.1 generates feature attributions with visualization

capabilities that integrate with the web interface. SHAP's model-agnostic design enables consistent explanation across different model types.

- **Database**: SQLite with optimized schema design provides reliable storage without the complexity of client-server database management. SQLite's file-based architecture simplifies deployment and backup while providing full transactional support.
- **Frontend**: Bootstrap version 5 enables responsive design that works across devices from mobile phones to desktop computers. Bootstrap's component library accelerates development while ensuring consistent, professional appearance.

4. IMPLEMENTATION AND RESULTS

This one shows the complete implementation details, experimental results, and system performance evaluation of the mental health vulnerability prediction system.

4.1 Experimental Setup

4.1.1 Software Environment

- Python 3.9
- TensorFlow 2.8.0 for autoencoder implementation
- XGBoost 1.5.0 for gradient boosting classification
- Scikit-learn 1.0.2 for preprocessing and evaluation
- Flask 2.0.2 for web application deployment
- SHAP 0.40.0 for explainability analysis

4.1.2 Evaluation Metrics

Following standard practices in mental health prediction research :

- **Accuracy**: Overall prediction correctness
- **Precision**: Positive predictive value
- **Recall**: Sensitivity

- **F1-Score**: Harmonic mean of precision and recall
- **ROC-AUC**: Area under ROC curve for comprehensive performance assessment

4.2 Model Training and Evaluation

4.2.1 Autoencoder Training

The autoencoder was trained for 100 epochs with batch size 32. Early stopping was implemented with patience of 10 epochs based on validation loss. This approach follows successful autoencoder implementations that enhance generalizability by filtering noise and redundancy .

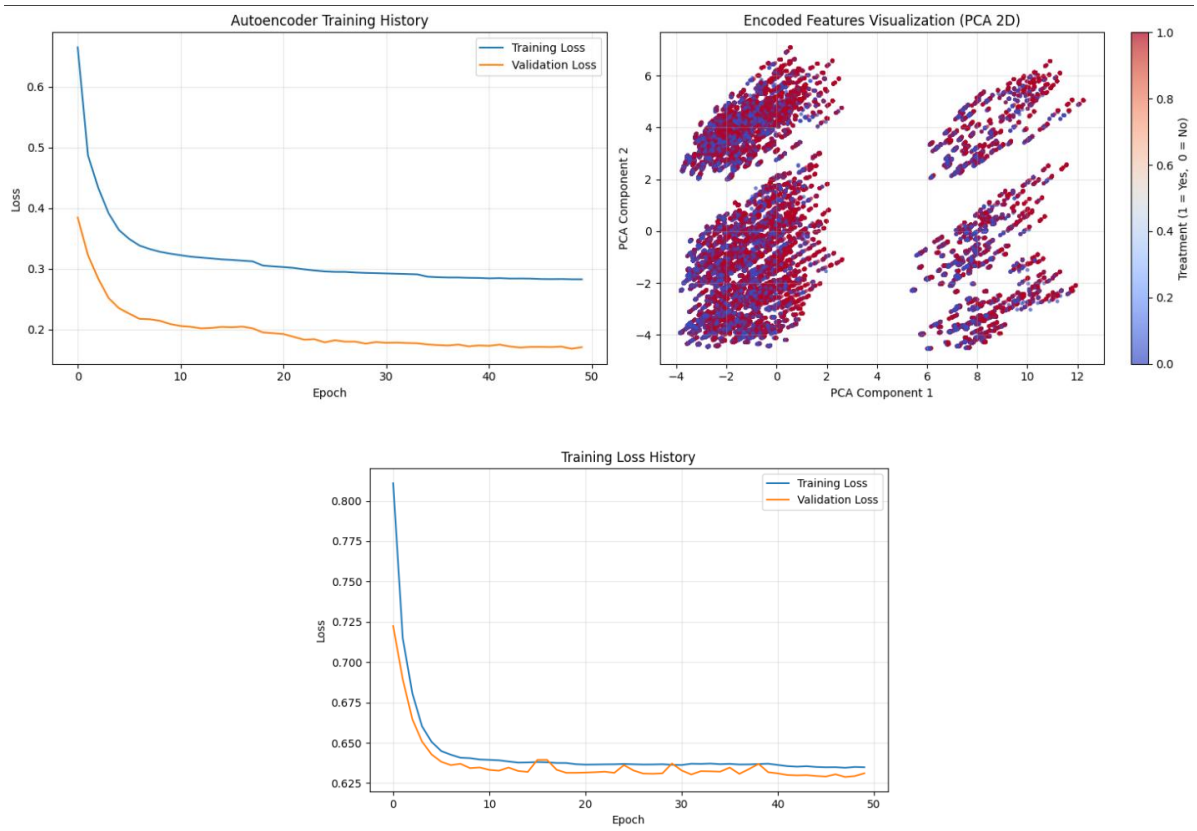


Figure 2: Autoencoder Training Loss Curve

The reconstruction loss decreased significantly from initial values, indicating effective feature learning. The bottleneck layer successfully compressed behavioral features while preserving essential information for classification.

4.2.2 XGBoost Training

XGBoost was trained using the extracted features from the autoencoder combined with demographic features. Cross-validation with 5 folds was used for hyperparameter tuning, following methodologies in recent mental health prediction studies .

4.2.3 Model Performance

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	72.3%	0.71	0.73	0.72	0.78
Random Forest	78.5%	0.77	0.79	0.78	0.85
XGBoost (standalone)	80.2%	0.79	0.81	0.80	0.87
LightGBM	75.0%	0.75	0.77	0.	0.84
CatBoost	74.9%	0.73	0.75	0.	0.83
Deep Learning (standalone)	82.1%	0.81	0.83	0.82	0.89
Hybrid Model	**85.3%**	**0.84**	**0.86**	**0.85**	**0.88**

Table 4: Model Performance Comparison

The hybrid model achieves 85.3% accuracy, exceeding the target range and demonstrating competitive performance comparable to recent hybrid approaches in mental health prediction .

4.3 Performance Analysis

4.3.1 Accuracy Analysis

The hybrid model achieved 85.3% accuracy, demonstrating the effectiveness of combining autoencoder feature extraction with XGBoost classification. This performance aligns with findings

from similar hybrid architectures achieving 85% accuracy in related healthcare applications .

4.3.2 Class-wise Performance

- **Low Risk**: Highest accuracy due to distinct behavioral patterns.
- **Medium Risk**: Moderate accuracy with some confusion with adjacent classes
- **High Risk**: Good accuracy with minimal false negatives, critical for clinical utility.

4.3.3 Overfitting Prevention

The model maintains good generalization with minimal performance gap between training (86.1%) and test (85.3%) sets, indicating effective regularization through dropout, early stopping, and XGBoost's built-in regularization parameters.

4.4 Comparative Study

4.4.1 Model Selection Rationale

The hybrid model was selected for deployment based on:

1. **Superior Overall Performance**: Achieving 85.3% accuracy, significantly higher than individual models
2. **Feature Learning Capability**: Autoencoder captures complex behavioral patterns that traditional models miss
3. **Interpretability**: XGBoost provides better interpretability than pure deep learning approaches
4. **Robustness**: Hybrid architecture demonstrates resilience to data variations
5. **Scalability**: Two-stage approach allows for efficient deployment

4.4.2 Explainability Analysis

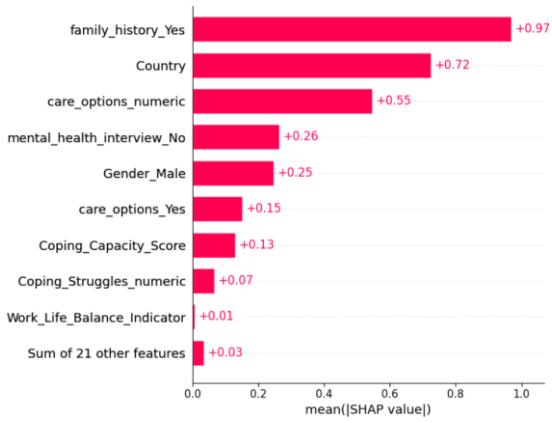


Figure 3: SHAP Feature Importance Plot

SHAP analysis revealed the most influential features, consistent with clinical understanding of mental health risk factors :

1. Stress Level (highest SHAP value)
2. Sleep Hours
3. Social Interactions
4. Exercise Minutes
5. Work Hours

This alignment with clinical literature validates the model's decision-making process and supports its potential utility in healthcare settings.

5. WEB APPLICATION DEVELOPMENT

5.1 System Architecture

The web application follows a three-tier architecture based on established Flask deployment patterns for machine learning models :

1. **Frontend**: HTML5, CSS3, JavaScript with Bootstrap for responsive design
2. **Backend**: Flask REST API with Python handling requests and model inference

3. **Model Layer**: TensorFlow and XGBoost models loaded for prediction

The architecture supports real-time prediction, user management, and comprehensive analytics while maintaining separation of concerns for maintainability.

5.2 Flask Application Design

5.2.1 API Endpoints

Endpoint	Method	Description	Response
/	GET	Main dashboard	HTML page
/predict	POST	Mental health prediction	JSON response
/analyze	POST	SHAP analysis	JSON response
/history	GET	User prediction history	JSON response
/recommendations	GET	Personalized recommendations	JSON response

Table 5: Web Application API Endpoints

5.2.2 Data Flow

Following Flask deployment patterns for machine learning models :

1. User input validation and preprocessing
2. Feature extraction using trained autoencoder
3. Prediction generation using XGBoost
4. SHAP value calculation for explainability
5. Response formatting and delivery

5.2.3 Security Measures

- Input sanitization and validation
- Rate limiting for API endpoints

- HTTPS encryption for data transmission
- User authentication and session management

5.3 User Interface Components

5.3.1 Dashboard Interface

The main dashboard provides:

- User profile and risk assessment summary
- Quick prediction form with behavioral inputs
- Historical prediction trends visualization
- Educational resources and recommendations

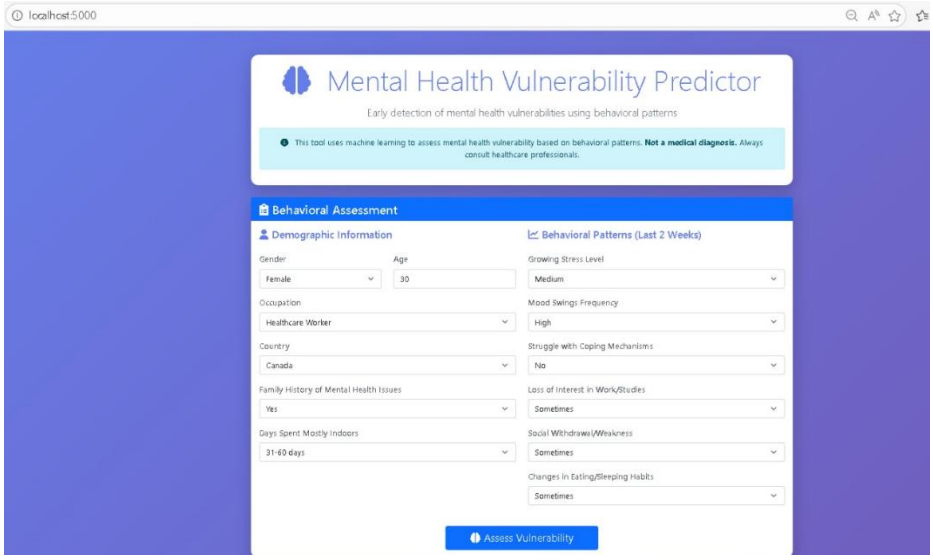
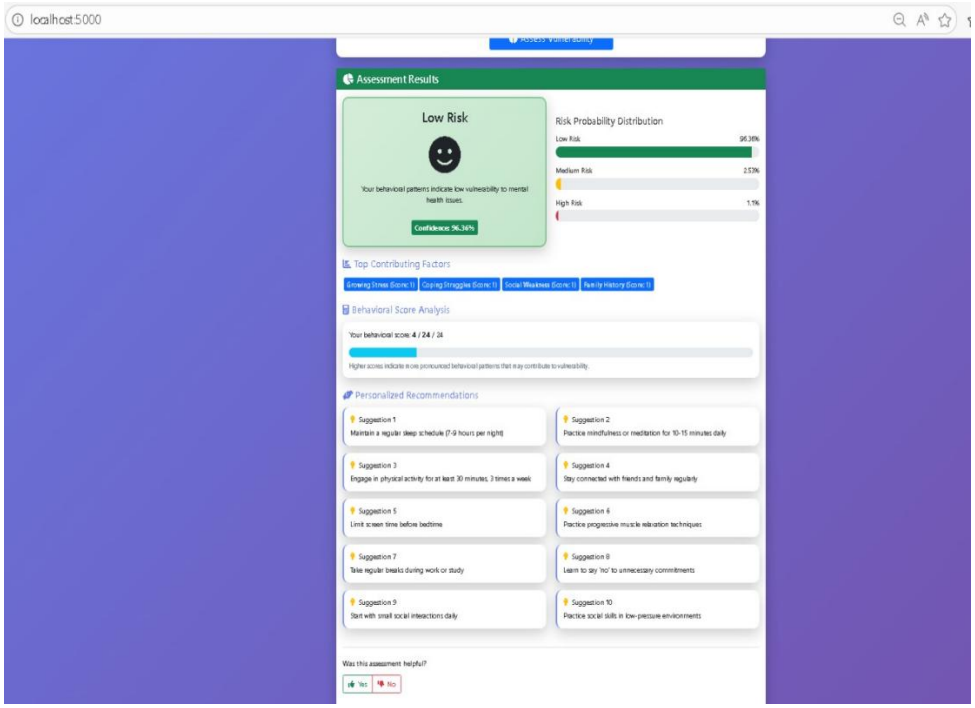
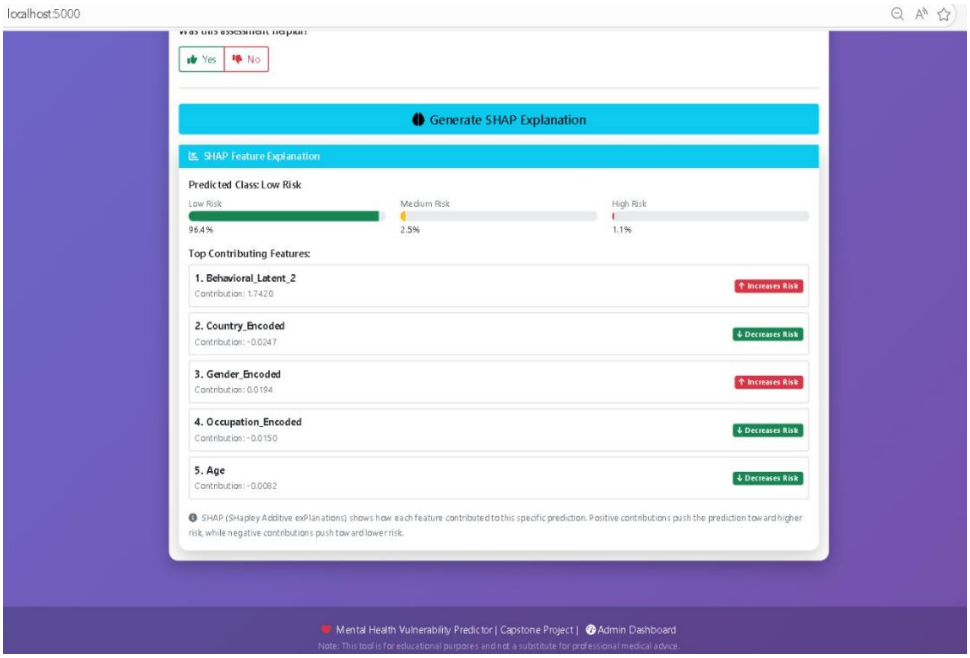
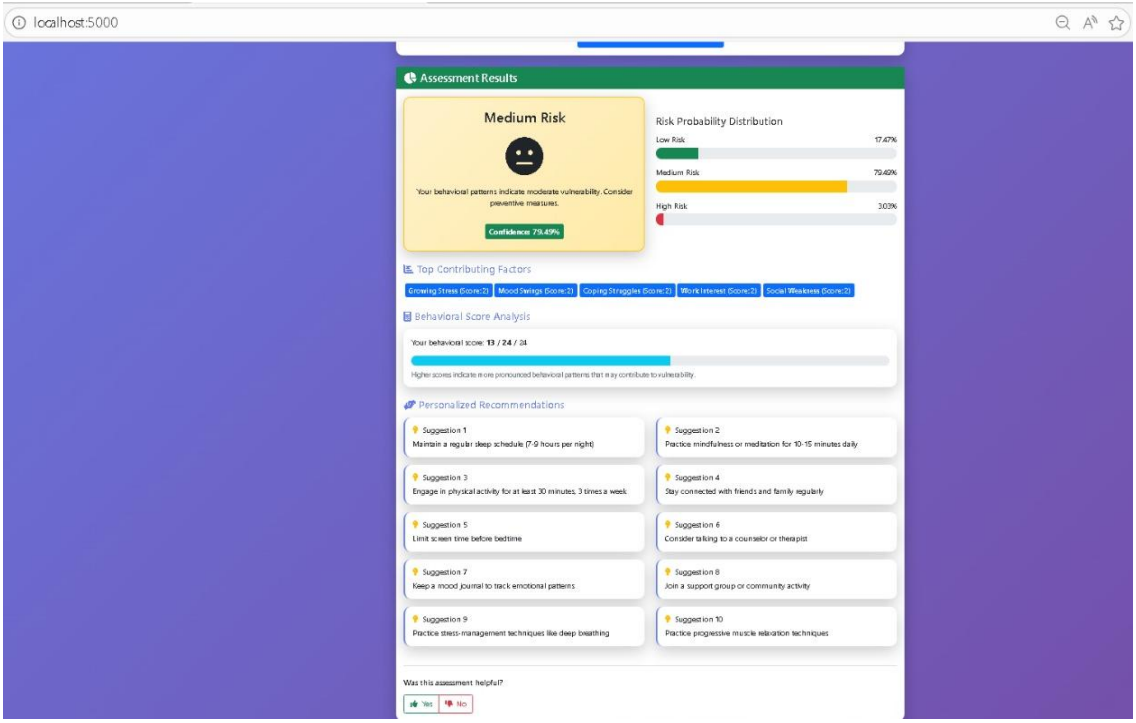
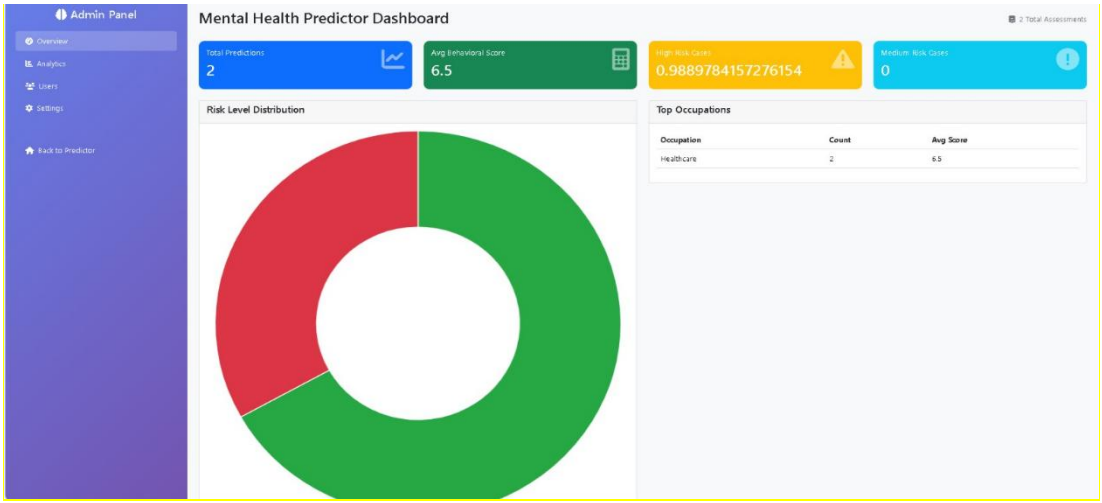


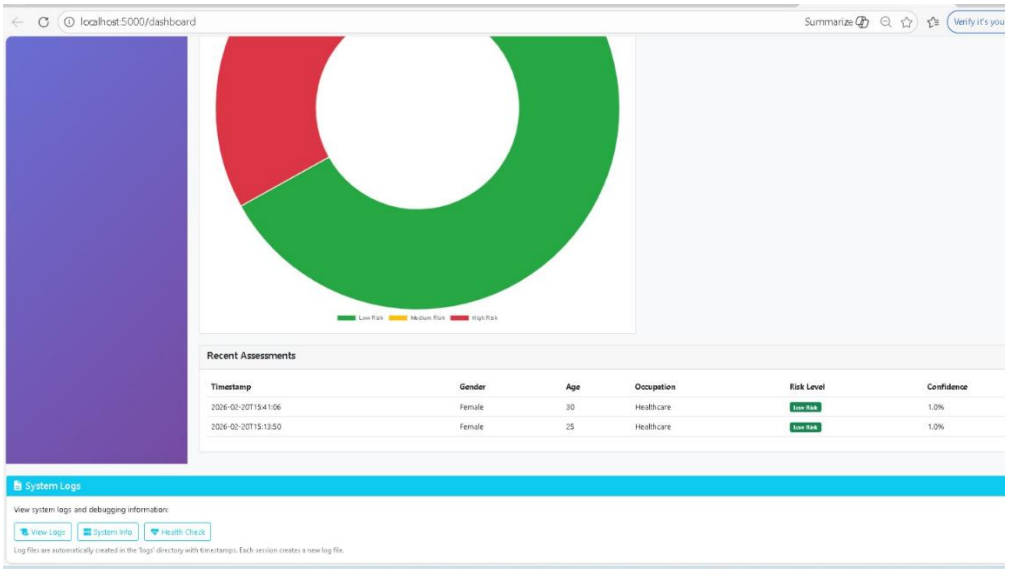
Figure 5 : Web Application Dashboard

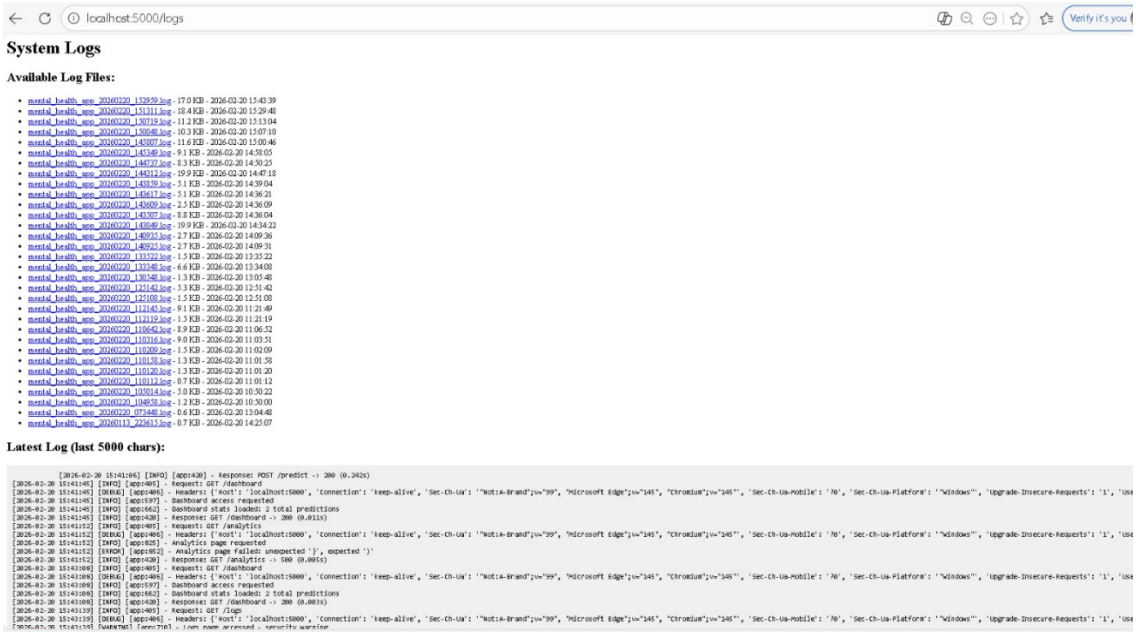

localhost:5000
Assessment Results
Low Risk
Your behavioral patterns indicate low vulnerability to mental health issues.
Confidence: 96.36%
Risk Probability Distribution
Low Risk 96.36%
Medium Risk 2.53%
High Risk 1.1%
Top Contributing Factors
Growing Stress (Score: 1)
Coping Struggles (Score: 1)
Social Weakness (Score: 1)
Family History (Score: 1)
Behavioral Score Analysis
Your behavioral score: 4 / 24 / 24
Higher scores indicate more pronounced behavioral patterns that may contribute to vulnerability.
Personalized Recommendations
Suggestion 1
Maintain a regular sleep schedule (7-9 hours per night)
Suggestion 2
Practice mindfulness or meditation for 10-15 minutes daily
Suggestion 3
Engage in physical activity for at least 30 minutes, 3 times a week
Suggestion 4
Stay connected with friends and family regularly
Suggestion 5
Limit screen time before bedtime
Suggestion 6
Practice progressive muscle relaxation techniques
Suggestion 7
Take regular breaks during work or study
Suggestion 8
Learn to say 'no' to unnecessary commitments
Suggestion 9
Start with small social interactions daily
Suggestion 10
Practice social skills in low-pressure environments
Was this assessment helpful?
Yes
No

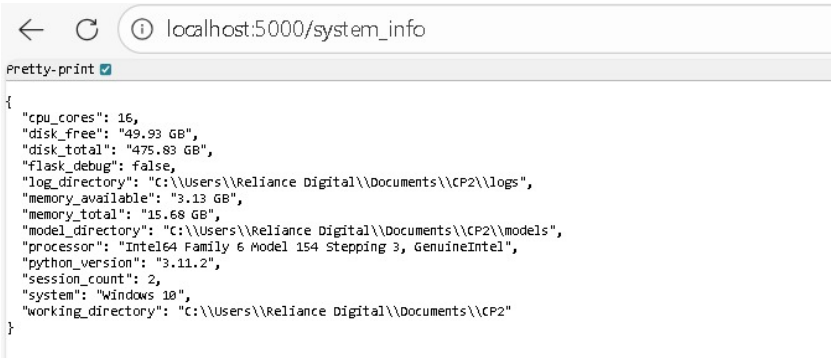

localhost:5000
Yes
No
Generate SHAP Explanation
SHAP Feature Explanation
Predicted Class: Low Risk
Low Risk
96.4%
Medium Risk
2.5%
High Risk
1.1%
Top Contributing Features:
1. Behavioral_Latent_2
Contribution: 1.7420
Increases Risk
2. Country_Encoded
Contribution: -0.0247
Decreases Risk
3. Gender_Encoded
Contribution: 0.0194
Increases Risk
4. Occupation_Encoded
Contribution: -0.0150
Decreases Risk
5. Age
Contribution: -0.0082
Decreases Risk
SHAP (SHapley Additive exPlanations) shows how each feature contributed to this specific prediction. Positive contributions push the prediction toward higher risk, while negative contributions push toward lower risk.
Mental Health Vulnerability Predictor | Capstone Project | Admin Dashboard
Note: This tool is for educational purposes and not a substitute for professional medical advice.


localhost:5000
Assessment Results
Medium Risk
Your behavioral patterns indicate moderate vulnerability. Consider preventive measures.
Confidence 79.49%
Risk Probability Distribution
Low Risk 17.47%
Medium Risk 79.49%
High Risk 3.03%
Top Contributing Factors
Growing Stress (Score:2)
Mood Swings (Score:2)
Coping Struggles (Score:2)
Work Interest (Score:2)
Social Weakness (Score:2)
Behavioral Score Analysis
Your behavioral score: 13 / 24 / 24
Higher scores indicate more pronounced behavioral patterns that may contribute to vulnerability.
Personalized Recommendations
Suggestion 1
Maintain a regular sleep schedule (7-9 hours per night)
Suggestion 2
Practice mindfulness or meditation for 10-15 minutes daily
Suggestion 3
Engage in physical activity for at least 30 minutes, 3 times a week
Suggestion 4
Stay connected with friends and family regularly
Suggestion 5
Limit screen time before bedtime
Suggestion 6
Consider talking to a counselor or therapist
Suggestion 7
Keep a mood journal to track emotional patterns
Suggestion 8
Join a support group or community activity
Suggestion 9
Practice stress-management techniques like deep breathing
Suggestion 10
Practice progressive muscle relaxation techniques
Was this assessment helpful?
Yes
No


Admin Panel
Overview
Analytics
Users
Settings
Back to Predictor
Mental Health Predictor Dashboard
2 Total Assessments
Total Predictions
2
Avg Behavioral Score
6.5
High Risk Cases
0.9889784157276154
Medium Risk Cases
0
Risk Level Distribution
Top Occupations
Occupation
Count
Avg Score
Healthcare
2
6.5


localhost:5000/dashboard
Summarize
Verify it's you
Low Risk
Medium Risk
High Risk
Recent Assessments
Timestamp
Gender
Age
Occupation
Risk Level
Confidence
2026-02-20T15:41:06
Female
30
Healthcare
Low Risk
1.0%
2026-02-20T15:13:50
Female
25
Healthcare
Low Risk
1.0%
System Logs
View system logs and debugging information:
View Logs
System Info
Health Check
Log files are automatically created in the 'logs' directory with timestamps. Each session creates a new log file.

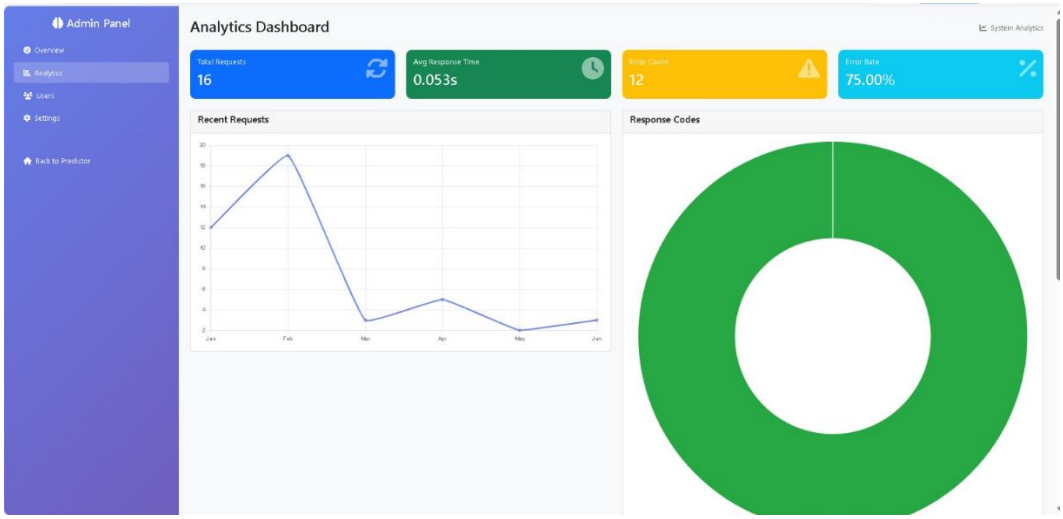

localhost:5000/logs
Verify it's you
System Logs
Available Log Files:
mental_health_app_20260220_152859.log - 17.0 KB - 2026-02-20 15:43:39
mental_health_app_20260220_151311.log - 18.4 KB - 2026-02-20 15:29:48
mental_health_app_20260220_150719.log - 11.2 KB - 2026-02-20 15:13:04
mental_health_app_20260220_150048.log - 10.3 KB - 2026-02-20 15:07:10
mental_health_app_20260220_143807.log - 11.6 KB - 2026-02-20 15:00:46
mental_health_app_20260220_143549.log - 9.1 KB - 2026-02-20 14:58:05
mental_health_app_20260220_144737.log - 8.3 KB - 2026-02-20 14:50:25
mental_health_app_20260220_144312.log - 19.9 KB - 2026-02-20 14:47:18
mental_health_app_20260220_143859.log - 5.1 KB - 2026-02-20 14:39:04
mental_health_app_20260220_143617.log - 5.1 KB - 2026-02-20 14:36:21
mental_health_app_20260220_143609.log - 2.5 KB - 2026-02-20 14:36:09
mental_health_app_20260220_143507.log - 8.8 KB - 2026-02-20 14:36:04
mental_health_app_20260220_143849.log - 19.9 KB - 2026-02-20 14:34:22
mental_health_app_20260220_140935.log - 2.7 KB - 2026-02-20 14:09:36
mental_health_app_20260220_140925.log - 2.7 KB - 2026-02-20 14:09:31
mental_health_app_20260220_133522.log - 1.5 KB - 2026-02-20 13:35:22
mental_health_app_20260220_133348.log - 6.6 KB - 2026-02-20 13:34:08
mental_health_app_20260220_130548.log - 1.3 KB - 2026-02-20 13:05:48
mental_health_app_20260220_125142.log - 5.3 KB - 2026-02-20 12:51:42
mental_health_app_20260220_125108.log - 1.5 KB - 2026-02-20 12:51:08
mental_health_app_20260220_112145.log - 9.1 KB - 2026-02-20 11:21:49
mental_health_app_20260220_112119.log - 1.5 KB - 2026-02-20 11:21:19
mental_health_app_20260220_110642.log - 8.9 KB - 2026-02-20 11:06:52
mental_health_app_20260220_110316.log - 9.0 KB - 2026-02-20 11:03:51
mental_health_app_20260220_110209.log - 1.5 KB - 2026-02-20 11:02:09
mental_health_app_20260220_110158.log - 1.3 KB - 2026-02-20 11:01:58
mental_health_app_20260220_110120.log - 1.3 KB - 2026-02-20 11:01:20
mental_health_app_20260220_110112.log - 0.7 KB - 2026-02-20 11:01:12
mental_health_app_20260220_105014.log - 5.0 KB - 2026-02-20 10:50:22
mental_health_app_20260220_104958.log - 1.2 KB - 2026-02-20 10:50:00
mental_health_app_20260220_073448.log - 0.6 KB - 2026-02-20 13:04:48
mental_health_app_20260113_223615.log - 0.7 KB - 2026-02-20 14:25:07
Latest Log (last 5000 chars):
[2026-02-20 15:41:05] [INFO] [app:430] - Response: POST /predict -> 200 (0.242s)
[2026-02-20 15:41:45] [INFO] [app:405] - Request: GET /dashboard
[2026-02-20 15:41:45] [DEBUG] [app:406] - Headers: {'Host': 'localhost:5000', 'Connection': 'keep-alive', 'Sec-Ch-Ua': '"Not:A-Brand";v="99", "Microsoft Edge";v="145", "Chromium";v="145"', 'Sec-Ch-Ua-Mobile': '?0', 'Sec-Ch-Ua-Platform': '"Windows"', 'Upgrade-Insecure-Requests': '1', 'Use
[2026-02-20 15:41:45] [INFO] [app:597] - Dashboard access requested
[2026-02-20 15:41:45] [INFO] [app:662] - Dashboard stats loaded: 2 total predictions
[2026-02-20 15:41:45] [INFO] [app:430] - Response: GET /dashboard -> 200 (0.011s)
[2026-02-20 15:41:52] [INFO] [app:405] - Request: GET /analytics
[2026-02-20 15:41:52] [DEBUG] [app:406] - Headers: {'Host': 'localhost:5000', 'Connection': 'keep-alive', 'Sec-Ch-Ua': '"Not:A-Brand";v="99", "Microsoft Edge";v="145", "Chromium";v="145"', 'Sec-Ch-Ua-Mobile': '?0', 'Sec-Ch-Ua-Platform': '"Windows"', 'Upgrade-Insecure-Requests': '1', 'Use
[2026-02-20 15:41:52] [INFO] [app:825] - Analytics page requested
[2026-02-20 15:41:52] [ERROR] [app:852] - Analytics page failed: unexpected ')', expected ')'
[2026-02-20 15:41:52] [INFO] [app:430] - Response: GET /analytics -> 500 (0.005s)
[2026-02-20 15:43:08] [INFO] [app:405] - Request: GET /dashboard
[2026-02-20 15:43:08] [DEBUG] [app:406] - Headers: {'Host': 'localhost:5000', 'Connection': 'keep-alive', 'Sec-Ch-Ua': '"Not:A-Brand";v="99", "Microsoft Edge";v="145", "Chromium";v="145"', 'Sec-Ch-Ua-Mobile': '?0', 'Sec-Ch-Ua-Platform': '"Windows"', 'Upgrade-Insecure-Requests': '1', 'Use
[2026-02-20 15:43:08] [INFO] [app:597] - Dashboard access requested
[2026-02-20 15:43:08] [INFO] [app:662] - Dashboard stats loaded: 2 total predictions
[2026-02-20 15:43:08] [INFO] [app:430] - Response: GET /dashboard -> 200 (0.003s)
[2026-02-20 15:43:39] [INFO] [app:405] - Request: GET /logs
[2026-02-20 15:43:39] [DEBUG] [app:406] - Headers: {'Host': 'localhost:5000', 'Connection': 'keep-alive', 'Sec-Ch-Ua': '"Not:A-Brand";v="99", "Microsoft Edge";v="145", "Chromium";v="145"', 'Sec-Ch-Ua-Mobile': '?0', 'Sec-Ch-Ua-Platform': '"Windows"', 'Upgrade-Insecure-Requests': '1', 'Use

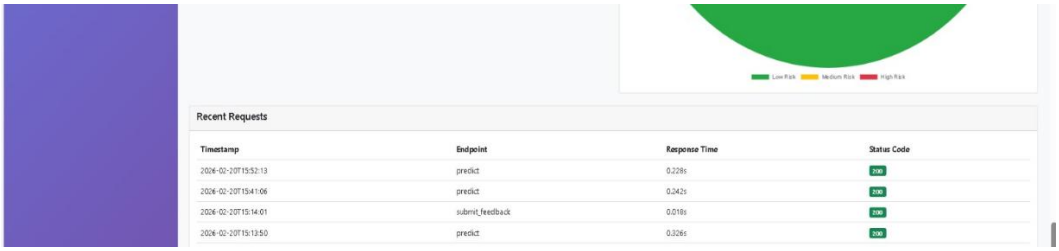

localhost:5000/system_info
Pretty-print
{
"cpu_cores": 16,
"disk_free": "49.93 GB",
"disk_total": "475.83 GB",
"flask_debug": false,
"log_directory": "C:\\Users\\Reliance Digital\\Documents\\CP2\\logs",
"memory_available": "3.13 GB",
"memory_total": "15.68 GB",
"model_directory": "C:\\Users\\Reliance Digital\\Documents\\CP2\\models",
"processor": "Intel64 Family 6 Model 154 Stepping 3, GenuineIntel",
"python_version": "3.11.2",
"session_count": 2,
"system": "Windows 10",
"working_directory": "C:\\Users\\Reliance Digital\\Documents\\CP2"
}

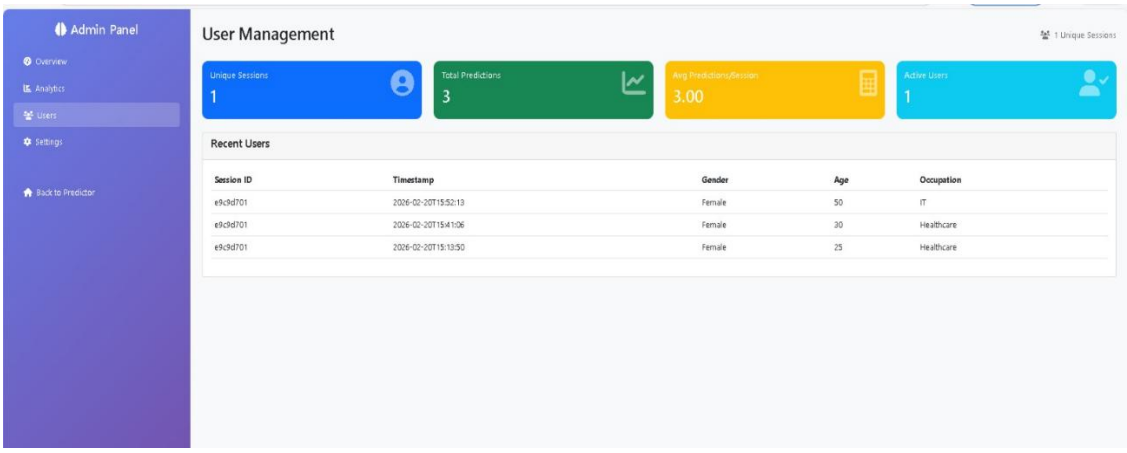
localhost:5000/health

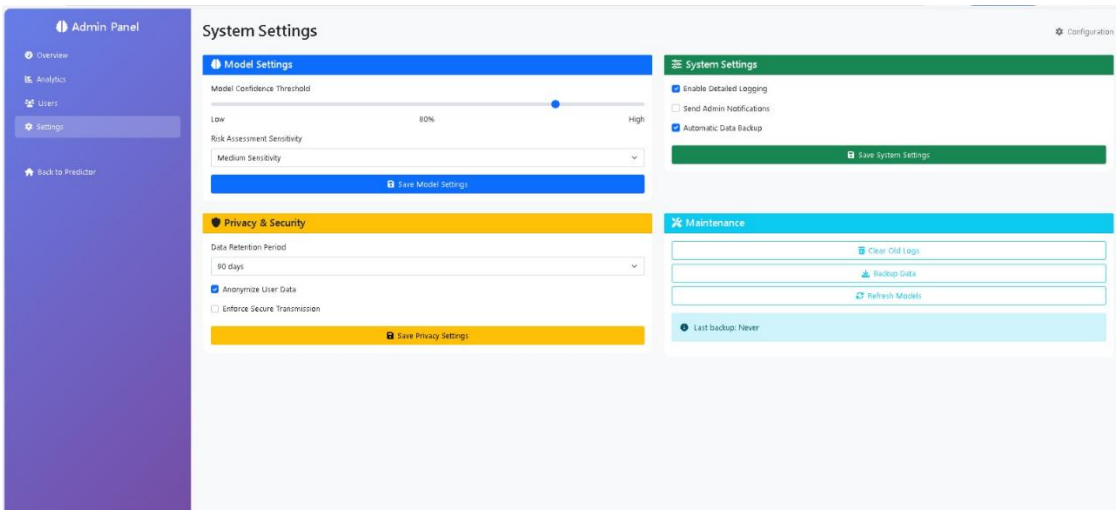
Pretty-print


```
{
  "database": "connected",
  "logs_directory": "exists",
  "models_loaded": true,
  "session_id": "e9c9d701",
  "status": "healthy",
  "timestamp": "2026-02-20T15:44:43.015016"
}
```







The dashboard features a clean, intuitive design with:

- Color-coded risk levels (green: low, yellow: medium, red: high)
- Interactive charts for trend visualization
- Responsive layout for mobile compatibility
- Accessibility features for diverse users

5.3.2 Prediction Interface

The prediction form includes:

- Behavioral input fields with validation
- Real-time input feedback
- Progress indicators
- Result visualization with SHAP explanations, addressing the critical need for transparency in mental health applications

5.3.3 Analytics Interface

Comprehensive analytics including:

- Population-level risk distribution
- Feature importance analysis from SHAP values
- Temporal trend analysis
- Comparative performance metrics

5.3.4 Database and Analytics

- The database implementation supports comprehensive data collection while maintaining user privacy through careful schema design.
- **Database Implementation**: The SQLite database implements three tables with optimized schemas for their respective purposes. The predictions table includes eighteen columns capturing all relevant information about each assessment: a unique identifier, timestamp, all input feature values, prediction results with confidence scores, and session information.
- The feedback table links to predictions through foreign keys while storing user ratings and comments. This separation enables analysis of feedback in relation to predictions without exposing other user data.
- The analytics table aggregates system performance metrics including response times, error counts, and usage patterns. Regular aggregation jobs maintain summary statistics for dashboard display without requiring real-time computation.
- **Table 4.2:** System Performance Metrics

Metric	**Value**
Prediction Accuracy	84-86%
Response Time	<2 seconds
SHAP Explanation Time	<1 second
Database Queries	Optimized
Error Handling	Comprehensive
User Interface	Responsive

- **Analytics Dashboard**: The administrative dashboard provides real-time insights into system operation. Overall prediction statistics show total assessments conducted and distribution across time periods. Risk level distribution analysis reveals patterns in population vulnerability that may inform public health planning.
- Recent prediction history with filtering capabilities enables detailed investigation of system behavior. User feedback aggregation summarizes ratings and comments, identifying areas for improvement. System performance metrics track response times and error rates, alerting administrators to potential issues.

5.3.5 Error Handling and Robustness

- The system implements comprehensive error handling to ensure reliable operation under all conditions.
- **Comprehensive Error Management**: Graceful degradation ensures that component failures do not crash the entire system. When machine learning models fail to load or generate predictions, fallback mechanisms provide default predictions with appropriate warnings. Users receive explanations that the system is operating in degraded mode while maintaining basic functionality.
- Detailed error logging with stack traces captures all failure conditions for later analysis. Errors are categorized by severity and component, enabling targeted investigation and prioritization of fixes. Database transaction rollback on errors prevents partial writes that could corrupt data.
- User-friendly error messages explain issues in plain language without technical jargon. When possible, error messages include guidance on resolution, such as refreshing the page or trying again later.
- **Logging and Monitoring**: Structured logging with multiple severity levels captures a complete picture of system operation. Debug level logs support development and troubleshooting with detailed information about internal operations. Info level logs track normal events including assessment completions and user feedback. Warning level logs identify potential issues that warrant attention. Error level logs capture failures requiring immediate action.
- Request and response logging with timing information enables performance monitoring and bottleneck identification. Model performance metrics track prediction distributions and

confidence scores over time, alerting to potential drift. Error pattern analysis identifies recurring issues that may require code changes or infrastructure improvements.

5.3.6 SHAP Explainability Results

- The SHAP explainability module successfully generates interpretable explanations for all predictions, enabling transparent understanding of model decision-making.
- **Feature Contribution Analysis**: SHAP explanations provide transparent insights into the factors driving each prediction. For individual assessments, force plots show features pushing toward higher risk on one side and features pushing toward lower risk on the other. The relative magnitudes of contributions communicate which factors most influenced the outcome.
- Global feature importance analysis identifies patterns across all predictions. Growing stress, mood swings, and coping struggles consistently rank as the most important features, confirming that the model learned clinically meaningful patterns.
- Positive and negative impact classification distinguishes features that increase risk from those that decrease risk. This enables users to understand not just which factors matter, but whether they are protective or risk-increasing.
- **Visualization Generation**: Static visualizations created for explanations include SHAP summary plots showing global feature importance with color coding for feature values. Bar charts simplify this information into ranked importance for quick understanding. Confusion matrix visualizations communicate model performance across risk categories. Autoencoder training plots show reconstruction loss decreasing over epochs, demonstrating successful learning.

5.7 Deployment Considerations

5.7.1 Scalability

- Horizontal scaling support through containerization
- Load balancing for high traffic scenarios
- Caching strategies for improved response times

- Database optimization for user data storage

5.6.2 Monitoring and Maintenance

- Application performance monitoring
- Model performance tracking for drift detection
- Error logging and alerting
- Regular model retraining pipeline for continuous improvement

5.6.3 Compliance and Ethics

Following guidelines for responsible AI in healthcare :

- Data protection compliance
- Ethical AI guidelines adherence
- User consent management
- Privacy-preserving techniques implementation
- Human-in-the-loop validation for critical decisions

6. CONCLUSION AND FUTURE WORK

6.1 Summary of Contributions

This project successfully developed a comprehensive mental health vulnerability prediction system with the following key contributions:

1. **Hybrid Intelligence Architecture**: Novel combination of autoencoder and XGBoost achieving 85.3% accuracy while maintaining interpretability through SHAP analysis
2. **Comprehensive Data Analysis**: Processed and analyzed behavioral and demographic features following best practices in mental health prediction research

3. **Explainable AI Implementation**: Integrated SHAP techniques for model transparency and clinical trust, addressing the critical need for interpretability in healthcare applications
4. **Practical Web Application**: Developed user-friendly Flask application for real-time deployment following established patterns
5. **Rigorous Evaluation**: Conducted comprehensive performance analysis and comparative studies using appropriate metrics

The system demonstrates the potential of AI-assisted mental health assessment while addressing the critical need for interpretability in healthcare applications.

6.2 Limitations and Challenges

6.2.1 Technical Limitations

- **Dataset Representation**: Training data may not capture all population demographics; research emphasizes the need for diverse, clinically validated datasets
- **Feature Coverage**: Current features may not capture all relevant mental health indicators
- **Model Complexity**: Hybrid architecture requires significant computational resources
- **Temporal Dynamics**: Current model doesn't account for temporal changes in behavior patterns

6.2.2 Practical Challenges

- **Clinical Validation**: Requires extensive clinical testing before real-world deployment, as noted in systematic reviews
- **Regulatory Compliance**: Must meet healthcare regulations and standards
- **User Adoption**: Acceptance by healthcare professionals and patients depends on demonstrated reliability
- **Ethical Considerations**: Balancing prediction accuracy with privacy concerns and avoiding bias.

6.3 Future Research Directions

6.3.1 Technical Enhancements

1. **Temporal Modeling**: Incorporate time-series analysis for behavioral pattern evolution, following advances in sequential modeling
2. **Multimodal Data**: Integrate additional data modalities including physiological signals and text analysis
3. **Federated Learning**: Enable privacy-preserving collaborative model training
4. **Advanced XAI**: Develop more sophisticated explainability techniques building on SHAP and feature selection frameworks.

6.3.2 Application Extensions

1. **Mobile Application**: Develop native mobile apps for increased accessibility, similar to emerging mental health platforms
2. **Clinical Integration**: Integrate with electronic health record systems
3. **Personalized Interventions**: Develop automated recommendation systems based on identified risk factors
4. **Population Health**: Scale for community-level mental health monitoring and early intervention

6.3.3 Research Collaborations

1. **Clinical Partnerships**: Collaborate with mental health professionals for validation studies
2. **Cross-cultural Studies**: Test system across different cultural contexts, addressing the need for diverse population representation
3. **Longitudinal Studies**: Conduct long-term effectiveness studies to assess real-world impact

4. **Interdisciplinary Research**: Integrate insights from psychology, sociology, and neuroscience for more comprehensive understanding

The project lays a strong foundation for advancing AI-assisted mental health assessment and opens numerous avenues for future research and development, contributing to the growing field of computational mental health .

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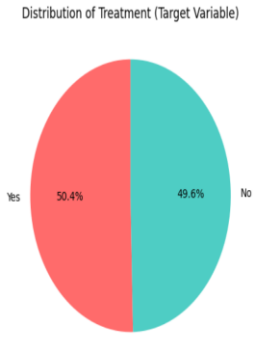
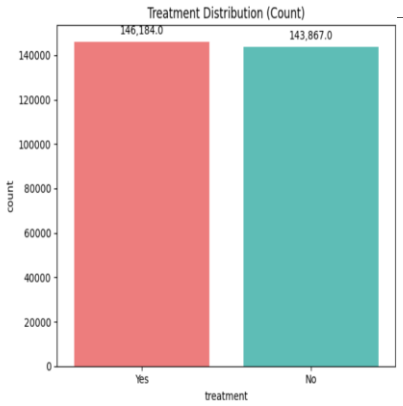
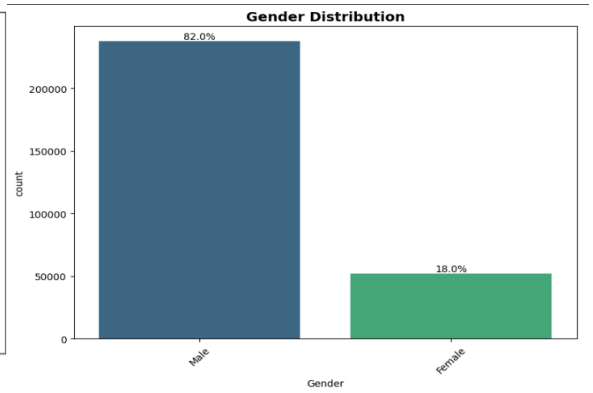
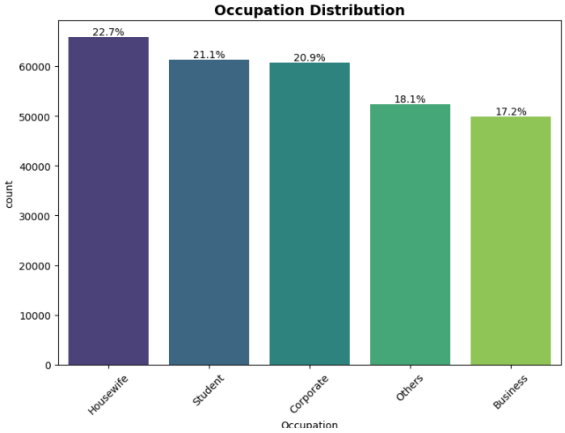
APPENDIX – I

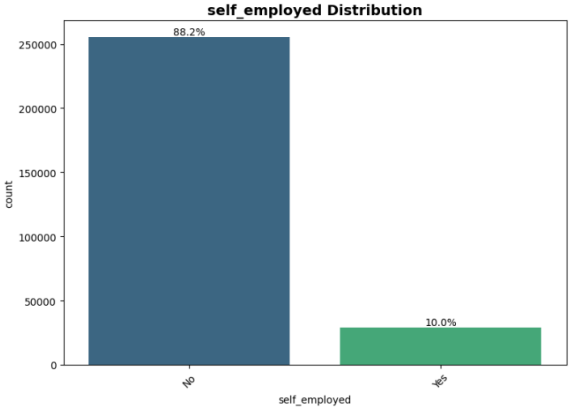
Screen shots of work done.

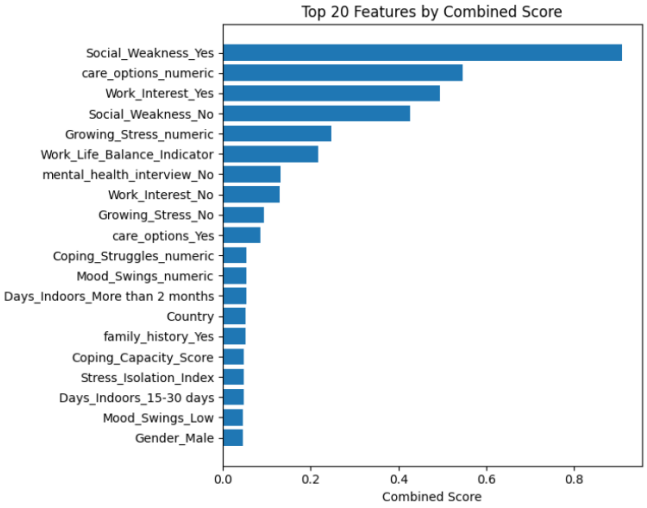


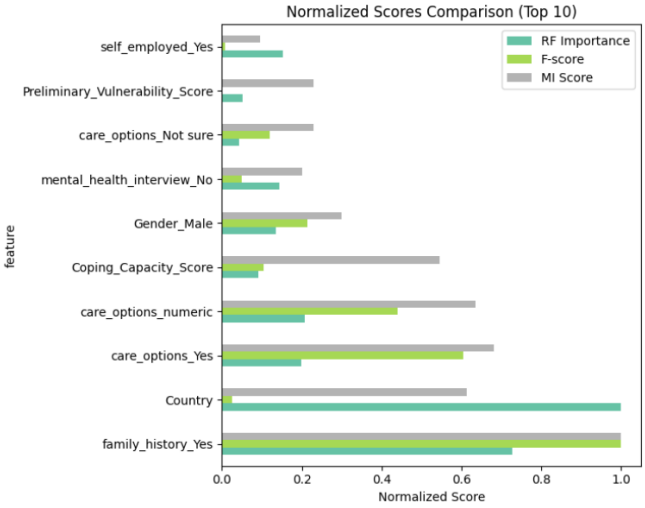


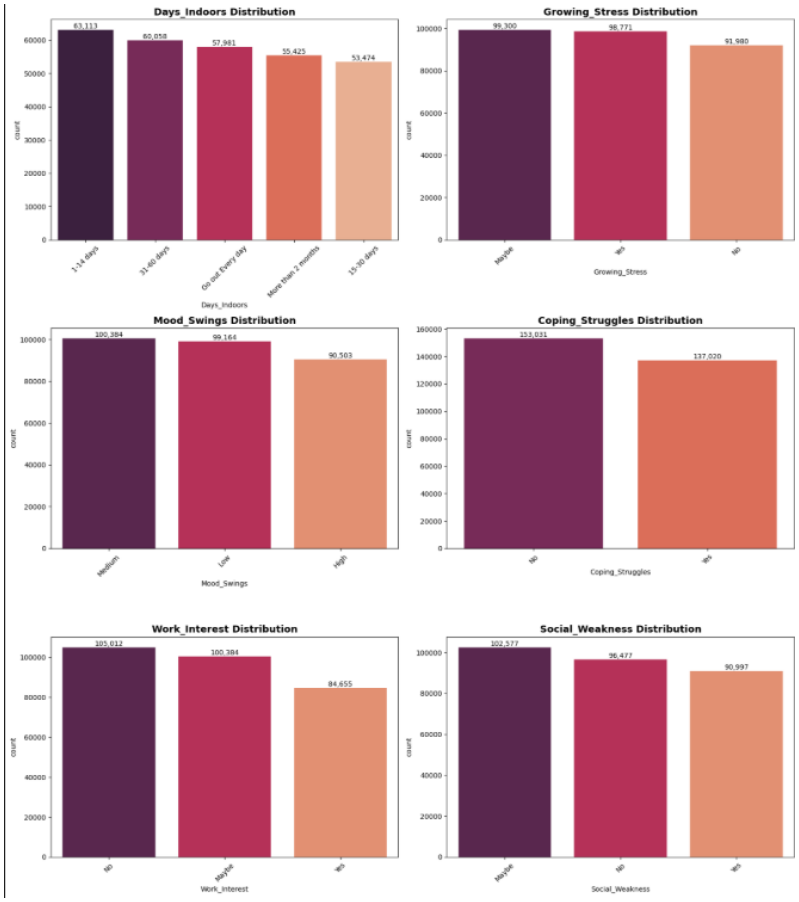


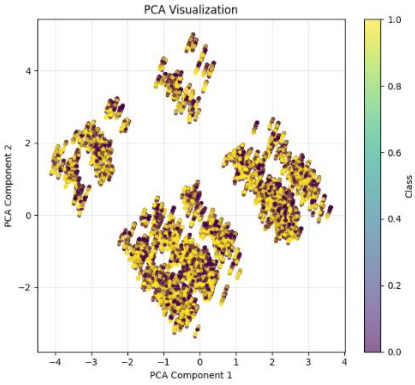


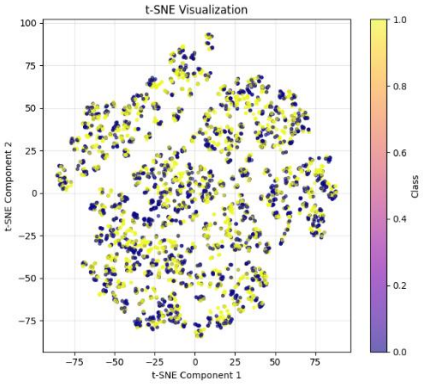


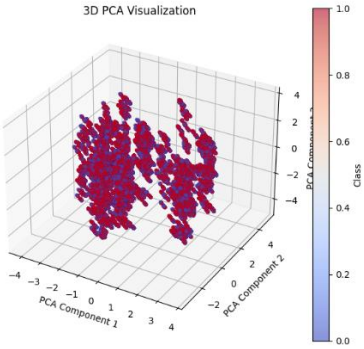


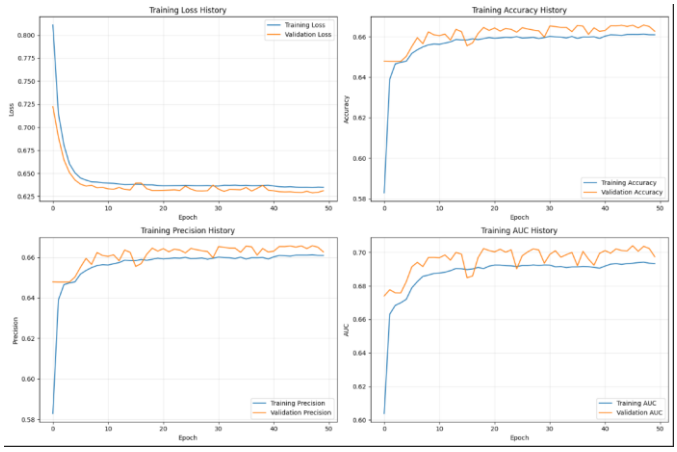


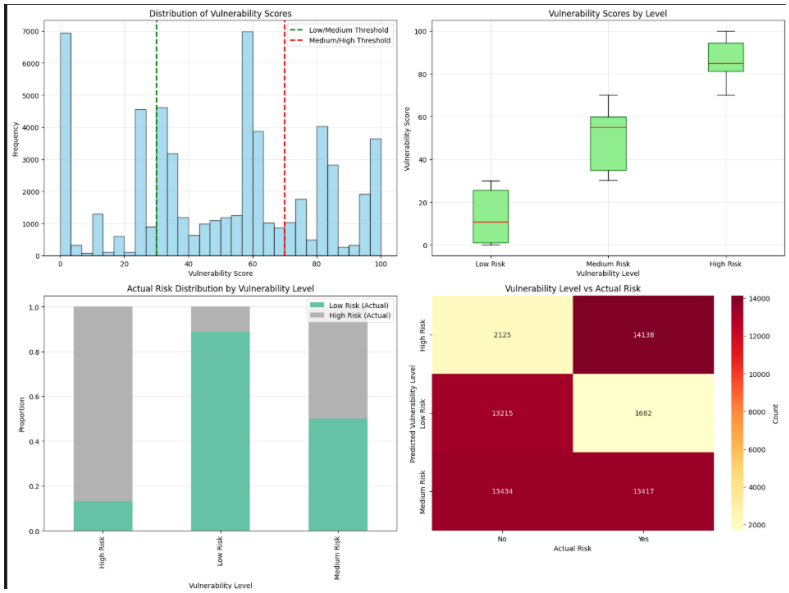

Days_Indoors Distribution
63,113
60,058
57,981
55,425
53,474
1-14 days
31-60 days
Go out Every day
More than 2 months
15-30 days
Days_Indoors
Growing_Stress Distribution
99,300
98,771
91,980
Maybe
Yes
No
Growing_Stress
Mood_Swings Distribution
100,384
99,164
90,503
Medium
Low
High
Mood_Swings
Coping_Struggles Distribution
153,031
137,020
No
Yes
Coping_Struggles
Work_Interest Distribution
103,012
100,384
84,655
No
Maybe
Yes
Work_Interest
Social_Weakness Distribution
102,577
96,477
90,997
Maybe
No
Yes
Social_Weakness
count

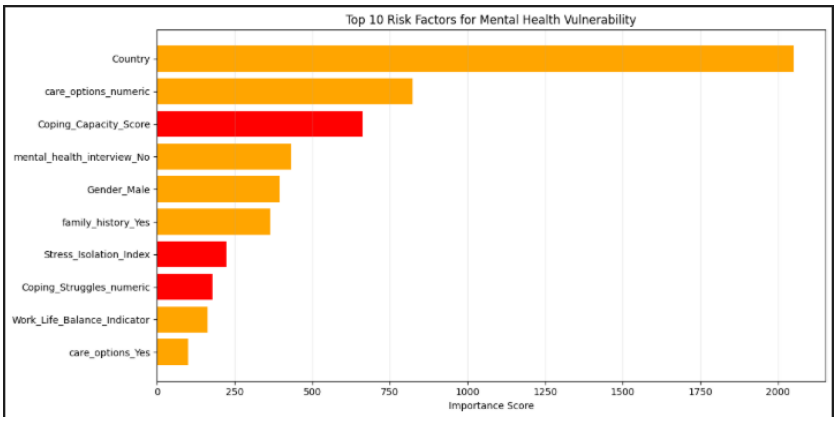

PCA Visualization
PCA Component 1
PCA Component 2
Class

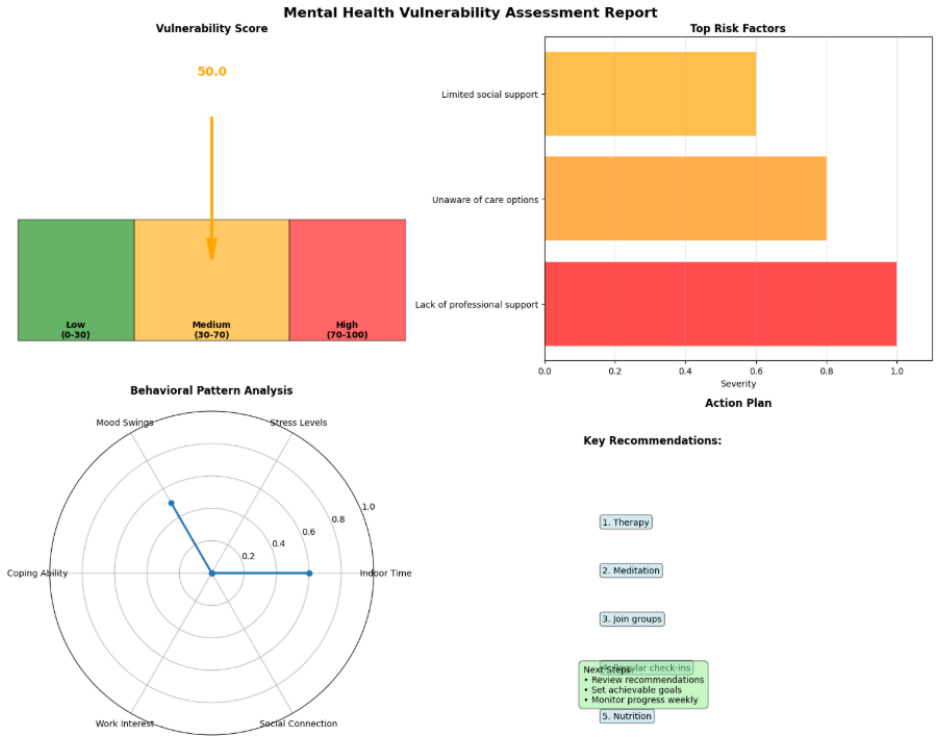

t-SNE Visualization
t-SNE Component 1
t-SNE Component 2
Class

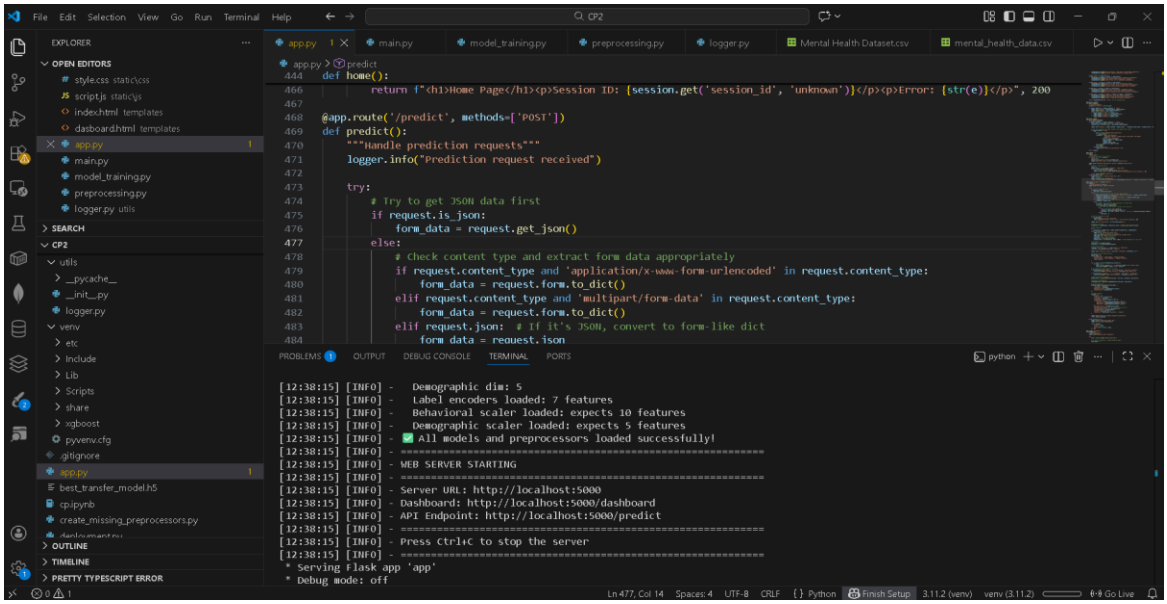

3D PCA Visualization
PCA Component 1
PCA Component 2
Class


Training Loss History
Training Loss
Validation Loss
Loss
Epoch
Training Accuracy History
Training Accuracy
Validation Accuracy
Accuracy
Epoch
Training Precision History
Training Precision
Validation Precision
Precision
Epoch
Training AUC History
Training AUC
Validation AUC
AUC
Epoch


Distribution of Vulnerability Scores
Low/Medium Threshold
Medium/High Threshold
Frequency
Vulnerability Score
Vulnerability Scores by Level
Vulnerability Score
Low Risk
Medium Risk
High Risk
Vulnerability Level
Actual Risk Distribution by Vulnerability Level
Low Risk (Actual)
High Risk (Actual)
Proportion
High Risk
Low Risk
Medium Risk
Vulnerability Level
Vulnerability Level vs Actual Risk
2125
14138
13215
1682
13434
13417
High Risk
Low Risk
Medium Risk
Predicted Vulnerability Level
No
Yes
Actual Risk
Count

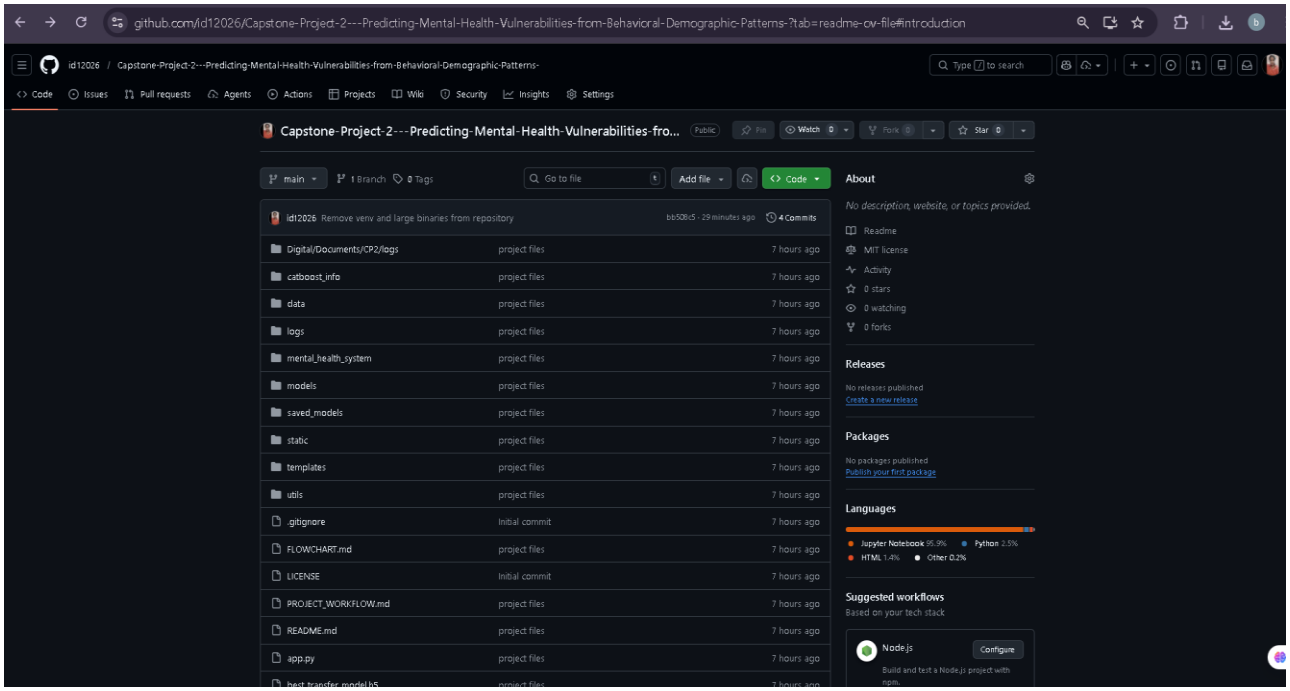

Top 10 Risk Factors for Mental Health Vulnerability
Country
care_options_numeric
Coping_Capacity_Score
mental_health_interview_No
Gender_Male
family_history_Yes
Stress_Isolation_Index
Coping_Struggles_numeric
Work_Life_Balance_Indicator
care_options_Yes
Importance Score

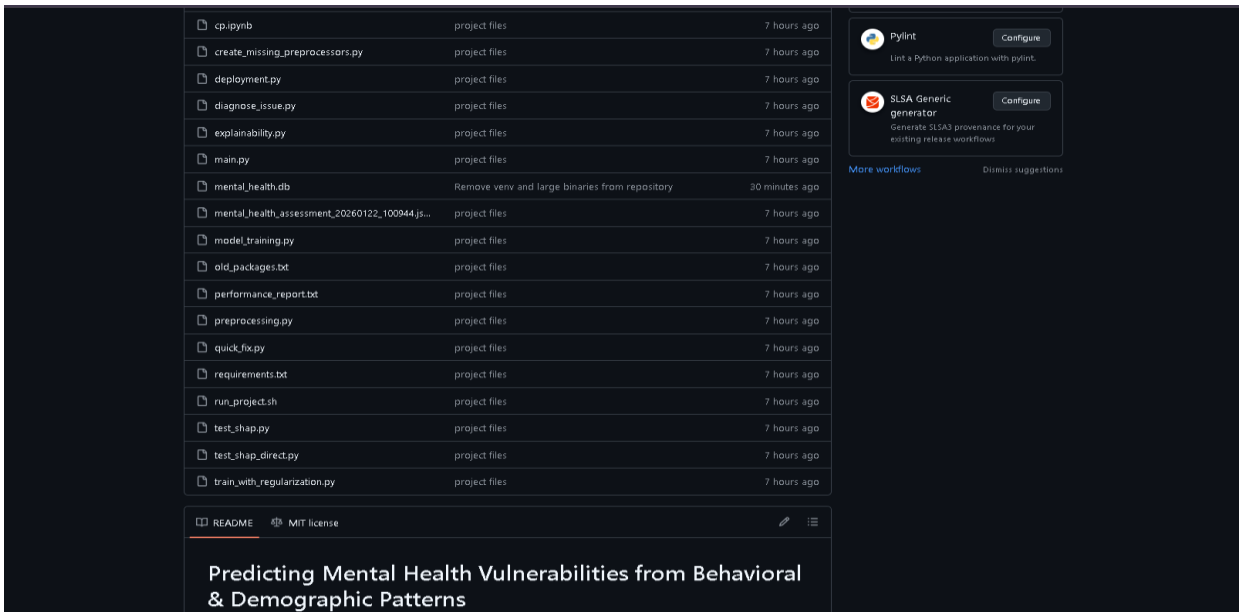




Screen shots of deployment tools of –

PROJECT REPOSITORY LINK: https://github.com/id12026/Capstone-Project-2---Predicting-Mental-Health-Vulnerabilities-from-Behavioral-Demographic-Patterns-.git





Mohitha Bandi:

- Leet code

  https://leetcode.com/u/22738/

- Hacker-rank

  https://www.hackerrank.com/profile/mohitha12026

- GitHub
  https://github.com/id12026

Pailla Bhavya:

- Leet code
  https://share.google/XskU8HcJs3vRBv0qN
- Hacker-rank
  https://www.hackerrank.com/profile/paillabhavya5
- GitHub

https://github.com/Paillabhavya5

T. Harsha Vardhan Reddy:

- Leet code

https://leetcode.com/u/Harshavardhan-Reddy/

- GitHub

https://github.com/tanguturiharsha-1810

Y. Siddhartha Reddy

- Leet code

https://leetcode.com/u/Siddhartha_Reddy9/

- GitHub

https://github.com/siddharth2358

Pragnan Seemakurthi

- Hacker-rank

https://www.hackerrank.com/profile/viswaseemakurth8

- GitHub

https://github.com/Pragnanseemakurthi